Learning Chemical Grammar: Dynamic Tokenization for SMILES with Hierarchical Networks

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Abstract

Tokenization fundamentally shapes how machine learning models represent chemical structures. While static tokenizers like SMILES Pair Encoding use predefined subword vocabularies, we investigate *dynamic tokenization* using Hierarchical Networks (H-Net), which learn byte-level chunking patterns directly from data. We train 350M-parameter H-Net models on polymeric (PI1M) and molecular (MOSES) SMILES datasets, systematically analyzing how dataset nature, training context, training duration, and architecture affect learned tokenization. Our study reveals three key findings: (1) **Dataset specificity matters**—H-Net learns distinct "chemical vocabularies" with only 30% token overlap between polymer and molecular datasets, analogous to language-specific patterns in NLP; (2) **Dynamic tokenization improves with training**—extended training yields 63% more unique tokens and 23% higher efficiency, achieving bits-per-byte of 0.64; (3) **Learned representations transfer**—H-Net embeddings outperform RDKit descriptors on blood-brain barrier penetration classification (AUC 0.95 vs. 0.93) and achieve competitive polymer glass transition temperature prediction. We establish dynamic tokenization as a promising paradigm for chemical representation learning.

1. Introduction

Chemical machine learning increasingly relies on string-based molecular representations, with SMILES (Simplified Molecular Input Line Entry System) (Weininger, 1988) serving as the dominant format. However, the *tokenization* of these chemical strings—how they are segmented into discrete units for model consumption—fundamentally constrains what patterns models can learn.

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Current approaches employ *static* tokenization strategies: character-level tokenization treats each symbol independently, losing chemical semantics; atom-level tokenization uses fixed chemical vocabularies; and subword methods like SmilesPE (Li & Fourches, 2021) apply BPE-style algorithms trained on large chemical corpora. While these methods provide standardized vocabularies, they impose uniform tokenization regardless of the specific chemical domain—the same tokens are used for drug-like molecules, polymers, and natural products alike.

This uniform approach may be suboptimal. In natural language processing, models trained on multilingual data learn language-specific tokenization patterns; Mandarin and English text yield fundamentally different token distributions even with shared architectures (Hwang et al., 2025). We hypothesize that chemistry exhibits analogous domain specificity: polymeric structures, with their repeating units and complex topologies, may benefit from different tokenization than small drug-like molecules.

To investigate this hypothesis, we apply *dynamic tokenization* using Hierarchical Networks (H-Net) (Hwang et al., 2025), which learn byte-level chunking patterns directly from data through differentiable boundary prediction. Unlike static tokenizers, H-Net adapts its tokenization to dataset characteristics, potentially discovering domain-specific chemical "grammars."

Contributions. We present the first systematic study of dynamic tokenization for chemical SMILES strings:

1. We train and analyze 8 H-Net models (350M parameters each) on polymeric and molecular datasets, systematically varying dataset nature, concatenation strategy, training duration, and architecture.

2. We develop a comprehensive metrics suite including token statistics, Jaccard similarity, KL divergence, breakpoint analysis, and bits-per-byte compression to characterize tokenization behavior.

3. We demonstrate that H-Net learns dataset-specific vocabularies with only 30% token overlap between polymer and molecular domains.

4. We validate that H-Net embeddings serve as effective chemical featurizers, outperforming traditional descriptors on classification tasks.

2. Related Work

Chemical String Representations. SMILES (Weininger, 1988) remains the dominant chemical string format despite known limitations including non-uniqueness and arbitrary atom ordering. SELFIES (Krenn et al., 2020) guarantees syntactic validity but produces longer strings. For polymers, PSMILES (Audus & de Pablo, 2017) extends SMILES notation with attachment point markers to represent repeating units.

Chemical Tokenization. Character-level tokenization is simple but ignores chemical structure. Atom-level methods preserve chemical meaning but use fixed vocabularies. SmilesPE (Li & Fourches, 2021) applies pair encoding to learn subword vocabularies from ChEMBL, achieving good compression while maintaining chemical interpretability. However, all these approaches are *static*—they produce identical tokenization regardless of the downstream dataset or task.

Learned Tokenization. H-Net (Hwang et al., 2025) introduces hierarchical networks with learned dynamic chunking, demonstrating that byte-level models can learn optimal segmentation directly from data. Originally developed for natural language, H-Net showed that different languages induce different tokenization patterns. Byte-level transformers (Wang et al., 2022) and patch-based vision models (Dosovitskiy et al., 2020) demonstrate similar principles in other domains. We extend this paradigm to chemistry.

Chemical Language Models. ChemBERTa (Chithrananda et al., 2020) and related models apply masked language modeling to SMILES. MolGPT (Bagal et al., 2022) uses autoregressive generation. These models typically employ fixed tokenization, focusing on downstream task performance rather than tokenization analysis. Our work complements these approaches by examining how dynamic tokenization affects learned representations.

3. Methods

3.1. H-Net Architecture

We employ the H-Net architecture (Hwang et al., 2025), which combines state-space models (Mamba blocks) with Transformer layers to enable dynamic chunking at the byte level.

1-Stage Architecture. Our primary configuration uses layout `["m4", ["T22"], "m4"]`: 4 Mamba encoder blocks, 22 Transformer blocks with boundary prediction, and 4 Mamba decoder blocks. The model processes raw bytes (vocabulary size 256) and learns to predict chunk boundaries within the Transformer layers.

2-Stage Architecture. We also evaluate a hierarchical variant `["m4", ["T1m4", ["T22"], "m4T1"], "m4"]` that introduces two levels of chunking: Stage 0 groups bytes into initial chunks, and Stage 1 groups these chunks into "super-chunks."

Model Dimensions. All models use $d_{\text{model}} = [1024, 1536]$ for encoder/decoder and main network respectively, with 16 attention heads and rotary embeddings. Total parameters: $\sim$350M.

3.2. Datasets

PI1M (Polymeric). $\sim$1M polymeric SMILES (PSMILES) strings representing polymer repeat units with attachment point markers (`[*]`). Characterized by longer sequences and complex structural patterns.

MOSES (Molecular). Standard molecular SMILES dataset containing drug-like small molecules. Shorter sequences with simpler structural motifs.

3.3. Training Configuration

Loss Function. We use cross-entropy loss for next-byte prediction with load-balancing regularization:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + 0.01 \times \mathcal{L}_{\text{LB}} \tag{1}$$

where $\mathcal{L}_{\text{LB}}$ encourages uniform token selection across the routing module.

Optimization. AdamW optimizer with hierarchical learning rate modulation: outer stages use $3\times$ base LR for higher plasticity, while the core network uses $0.9\times$ base LR for stability. Base LR: 10^{-4}, gradient accumulation: 8 steps.

Concatenation. We optionally concatenate 10 SMILES strings per training example (separated by spaces) to provide longer context during training.

3.4. Experimental Matrix

We train models varying four factors:

- **Dataset**: PI1M (polymer) vs. MOSES (molecular)
- **Concatenation**: None vs. $10\times$ SMILES
- **Epochs**: 1, 5, or 22
- **Architecture**: 1-stage vs. 2-stage

This yields 8 trained models (6 1-stage + 2 2-stage), summarized in Table 1.

Table 1. Summary of trained H-Net models. All models use ∼350M parameters.

Model	Dataset	Concat	Epochs	Arch
PI1M-1ep	PI1M	10×	1	1-stg
PI1M-5ep	PI1M	10×	5	1-stg
PI1M-22ep	PI1M	10×	22	1-stg
PI1M-nocat	PI1M	None	5	1-stg
PI1M-2stg	PI1M	10×	5	2-stg
MOSES-5ep	MOSES	10×	5	1-stg
MOSES-nocat	MOSES	None	5	1-stg
MOSES-2stg	MOSES	10×	5	2-stg

3.5. Analysis Metrics

Token Statistics. Total tokens, unique tokens, tokens per SMILES, mean token length.

Vocabulary Overlap. Jaccard similarity between token vocabularies: $J(A, B) = |A \cap B|/|A \cup B|$.

Distribution Divergence. KL divergence between token frequency distributions.

Breakpoint Analysis. Character positions where tokenization boundaries occur.

Compression. Bits-per-byte ($BPB = \mathcal{L}_{\text{CE}}/\ln 2$) measures compression efficiency. Lower is better; random prediction yields $BPB = 8.0$.

3.6. Benchmark: SmilesPE

We compare against SmilesPE (Li & Fourches, 2021), an industry-standard tokenizer with vocabulary pre-trained on ChEMBL (∼30K tokens). We apply SmilesPE to both PI1M and MOSES datasets for direct comparison.

3.7. Property Prediction Evaluation

To assess representation quality, we extract H-Net hidden states (768-dim) using mean pooling and train XGBoost predictors with 5-fold cross-validation on:

- **Polymer**: Glass transition temperature (Tg), Mass attenuation coefficient (MAC)
- **Molecular**: Lipophilicity (LogD), Blood-brain barrier penetration (BBBP)

Baseline: RDKit Morgan fingerprints + 200 physicochemical descriptors (∼2248 dimensions).

4. Results

We analyze tokenization behavior on 10,000 SMILES per dataset, providing statistically robust characterization.

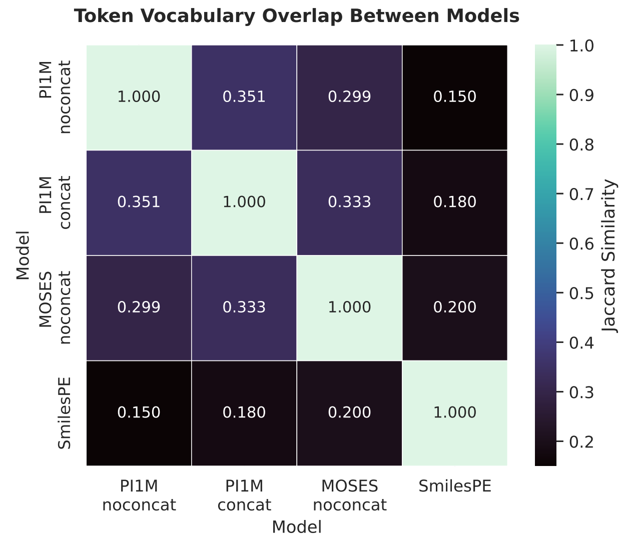

Figure 1. Token vocabulary overlap (Jaccard similarity) between models. Polymer and molecular models share only 30% of tokens, indicating dataset-specific vocabulary learning.

Table 2. Comparison of polymer (PI1M) vs. molecular (MOSES) tokenization under matched conditions (5 epochs, no concatenation).

Metric	Value	Interpretation
Token Overlap (Jaccard)	0.30	Distinct vocabularies
Breakpoint Overlap	0.57	Some shared motifs
KL Divergence	3.92	Large distribution gap
Tokens/SMILES (PI1M)	22.0	More complex
Tokens/SMILES (MOSES)	18.5	Simpler structures

4.1. Dataset Nature: Polymer vs. Molecular

Our central finding is that H-Net learns *distinct tokenization strategies* for polymeric versus molecular SMILES. Table 2 quantifies this difference.

Only 30% of tokens are shared between polymer and molecular models (Figure 1), comparable to the divergence between language-specific models in NLP. The KL divergence of 3.92 indicates substantially different frequency distributions. Polymer SMILES consistently require ∼20% more tokens, reflecting their inherent structural complexity.

Interpretation. Just as H-Net learns different tokens for Mandarin versus English (Hwang et al., 2025), it discovers domain-specific “chemical vocabularies”—byte patterns that efficiently represent polymer structures differ fundamentally from those optimal for drug-like molecules.

4.2. Concatenation Effect

Concatenating multiple SMILES during training provides longer context but could introduce noise from unrelated

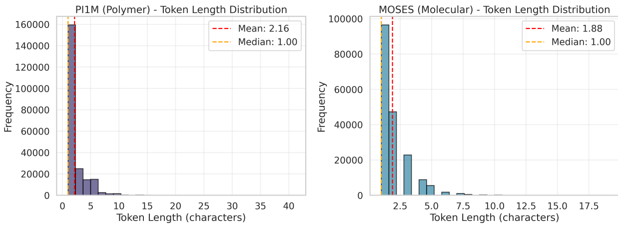

Figure 2. Token length distributions for polymer (PI1M) and molecular (MOSES) datasets. Polymer tokenization produces shorter tokens (2.2 vs. 2.4 characters on average), reflecting finer-grained segmentation.

Table 3. Effect of concatenation (10× SMILES) on tokenization stability.

Dataset	Token Overlap	Breakpoint Stability	Δ Unique
PI1M	0.35	0.76	+41%
MOSES	0.45	**0.96**	+99%

structures. Table 3 shows the effect.

Key finding. Molecular SMILES show remarkable stability under concatenation—96% of breakpoint positions are preserved. Polymer SMILES show more change (76% stability), likely because their already-long sequences benefit more from additional context.

Contrary to our initial hypothesis that polymers (being inherently concatenated structures) would be less affected, molecular SMILES actually showed greater stability. This suggests that the shorter, more stereotyped drug-like molecules have more consistent tokenization patterns.

4.3. Training Amount Effect

We examine how tokenization evolves with training using PI1M models at 1, 5, and 22 epochs (Table 4).

Extended training produces dramatic improvements: 63% more unique tokens, 23% fewer tokens per SMILES (higher efficiency), and 30% longer mean token length. Critically, breakpoint overlap between 1-epoch and 22-epoch models remains high (91%), indicating that training refines *what* tokens are learned while preserving *where* boundaries occur.

Compression. BPB improves from 0.83 (1 epoch) to 0.64 (22 epochs), far better than random prediction (8.0) and comparable to well-trained English language models ($\sim$1.0–1.5 BPB). This indicates H-Net learns meaningful chemical "grammar"—predictable patterns in functional groups, rings, and stereochemistry.

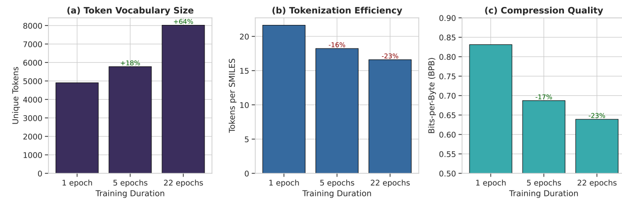

Figure 3. Effect of training duration on tokenization. Extended training produces (a) larger vocabularies, (b) fewer tokens per SMILES (higher efficiency), and (c) better compression (lower BPB).

Table 4. Effect of training duration on tokenization (PI1M, concatenated).

Epochs	Unique	Tok/SMILES	Tok Len	BPB
1	4,903	21.6	2.20	0.83
5	5,775	18.2	2.62	0.69
22	8,019	16.6	2.87	**0.64**
Δ	+63%	−23%	+30%	−23%

4.4. Architecture: 1-Stage vs. 2-Stage

The 2-stage architecture introduces hierarchical chunking. Table 5 compares compression performance.

2-stage provides marginal compression improvement (+0.2–0.4%) with comparable perplexity. The hierarchical structure adds interpretability (viewing tokenization at multiple levels) without significant performance impact. Notably, molecular SMILES benefit slightly more from hierarchy despite their simpler structure.

4.5. H-Net vs. SmilesPE

Table 6 compares H-Net against the industry-standard SmilesPE tokenizer.

The approaches embody different philosophies (Figure 4): H-Net learns fine-grained, byte-level patterns adaptively, producing larger vocabularies but more tokens per SMILES. SmilesPE uses predefined chemical subwords for efficient compression. These are *complementary* rather than competing—H-Net excels at discovering dataset-specific patterns, while SmilesPE provides interpretable, standardized tokens.

5. Application: Property Prediction

To validate that learned representations are useful, we evaluate H-Net embeddings for property prediction (Table 7).

Key findings:

- **Classification**: H-Net outperforms RDKit on BBBP by 2.3% AUC (0.950 vs. 0.927), demonstrating that

Table 5. Compression comparison: 1-stage vs. 2-stage architecture.

Dataset	1-Stage BPB	2-Stage BPB	Improvement
PI1M	0.687	0.686	+0.2%
MOSES	0.682	0.679	+0.4%

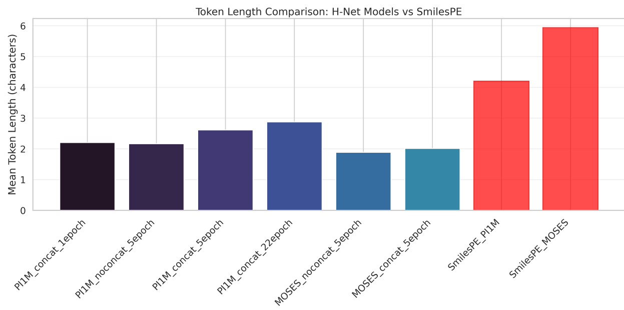

Figure 4. Token length distributions comparing H-Net (2–3 characters) and SmilesPE (4–6 characters). H-Net learns finer-grained, byte-level patterns while SmilesPE uses coarser chemical subwords.

learned representations capture decision-relevant structure.

- **Polymer Tg**: H-Net achieves 26.6°C MAE, competitive with RDKit (24.8°C) and comparable to specialized methods like Lieconv-Tg (Zhang et al., 2023) (24.4 K).
- **Precise regression**: RDKit outperforms on tasks requiring fine-grained numerical prediction (Lipophilicity, MAC).
- **Pooling matters**: Mean pooling is essential; CLS-token pooling fails catastrophically (AUC drops 26%).

These results suggest H-Net embeddings are complementary to traditional descriptors—useful for classification and polymer tasks, while RDKit excels at precise molecular regression.

6. Discussion

Why does dynamic tokenization work? Chemical strings contain domain-specific patterns—functional groups, ring systems, stereochemistry markers—that benefit from adaptive segmentation. Static tokenizers impose uniform boundaries that may not align with chemical semantics, while H-Net discovers dataset-relevant patterns through end-to-end learning.

Trade-offs. H-Net requires training (1–3 days on GPU) and produces less interpretable tokens than chemistry-aware methods. However, it offers adaptability to novel chemical domains, continuous improvement with more data, and competitive downstream performance. The choice depends on the application: use H-Net for domain-specific representation learning; use SmilesPE for standardized, interpretable tokenization.

Table 6. Comparison of H-Net vs. SmilesPE tokenization strategies.

Aspect	H-Net	SmilesPE
Token length	2–3 chars	4–6 chars
Vocabulary size	6–8K	1.6–2K
Tokens/SMILES	16–22	6–11
Adaptability	Dataset-specific	Fixed
Training	Required	Zero-shot

Table 7. Property prediction results comparing H-Net embeddings (mean pooling) vs. RDKit descriptors with XGBoost.

Task	Metric	RDKit	H-Net	Winner
BBBP	AUC↑	0.927	**0.950**	H-Net
Tg	MAE↓	**24.8°C**	26.6°C	RDKit
Lipophilicity	MAE↓	**0.494**	0.682	RDKit
MAC	MAE↓	**5.7e-5**	10.3e-5	RDKit

Limitations. Our study uses single model size (350M parameters); scaling behavior is unknown. Property prediction evaluation covers limited tasks. The byte-level approach increases sequence length compared to subword methods, impacting computational cost.

7. Conclusion

We presented the first systematic study of dynamic tokenization for chemical SMILES using H-Net. Our key findings are:

1. **Dataset specificity**: H-Net learns distinct tokenization strategies with only 30% vocabulary overlap between polymer and molecular domains.
2. **Training dynamics**: Extended training yields 63% more unique tokens and 23% higher efficiency without overfitting.
3. **Representation quality**: H-Net embeddings outperform RDKit on classification (BBBP) and achieve competitive polymer property prediction.

These results establish dynamic tokenization as a promising paradigm for chemical representation learning. Future work includes hybrid approaches combining H-Net's adaptability with chemistry-aware priors, interpretability analysis of learned tokens, and scaling to larger chemical corpora.

Impact Statement

This paper presents work whose goal is to advance machine learning methods for chemical representation. Potential applications include drug discovery and materials design, which carry both benefits (accelerated therapeutic development) and risks (dual-use concerns). We encourage responsible deployment with appropriate safety considerations.

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A. Extended Results

A.1. Complete Model Comparison

Table 8. Complete tokenization statistics for all models.

Model	Total Tokens	Unique Tokens	Tokens/SMILES	Mean Tok Len	BPB	PPL
PI1M (Polymer) Models						
PI1M-1ep	216,277	4,903	21.63	2.20	0.831	1.78
PI1M-5ep	182,298	5,775	18.23	2.62	0.687	1.61
PI1M-22ep	165,928	8,019	16.59	2.87	0.639	1.56
PI1M-nocat	220,423	4,094	22.04	2.17	0.670	1.59
PI1M-2stg	236,169	5,047	23.62	2.02	0.686	1.61
MOSES (Molecular) Models						
MOSES-5ep	172,964	6,183	17.30	2.01	0.682	1.60
MOSES-nocat	184,614	3,112	18.46	2.12	0.658	1.58
MOSES-2stg	186,361	3,799	18.64	1.87	0.679	1.60
SmilesPE Benchmark						
SmilesPE-PI1M	113,497	1,635	11.35	4.20	—	—
SmilesPE-MOSES	58,589	1,969	5.86	5.94	—	—

A.2. Property Prediction Details

Table 9. Complete property prediction results with standard deviations.

Model	Pooling	Tg MAE (°C)	MAC MAE	BBBP AUC
RDKit	—	24.8 ± 0.7	5.7e-5 $\pm$ 0.2e-5	0.927 ± 0.009
hnet_1ep_nocat	mean	26.6 ± 0.6	10.3e-5 $\pm$ 0.3e-5	—
hnet_5ep_nocat	mean	27.0 ± 0.5	10.6e-5 $\pm$ 0.3e-5	0.950 ± 0.002
hnet_5ep_cat	mean	26.9 ± 0.6	11.1e-5 $\pm$ 0.4e-5	0.949 ± 0.006
hnet_22ep_cat	mean	28.2 ± 0.5	11.6e-5 $\pm$ 0.3e-5	—
hnet_5ep_2stg	mean	27.4 ± 0.8	11.4e-5 $\pm$ 0.3e-5	0.950 ± 0.009
hnet_5ep_nocat	cls	67.9 ± 1.3	26.0e-5 $\pm$ 0.4e-5	0.706 ± 0.024

A.3. Training Configurations

All models were trained with the following configuration:

- **Optimizer**: AdamW with weight decay 0.1
- **Learning rate**: 1×10^{-4} base, with hierarchical modulation
- **Batch size**: 64 per GPU step, gradient accumulation 8 (effective 512)
- **Precision**: bfloat16
- **Hardware**: Single NVIDIA A100 GPU
- **Training time**: 1–3 days depending on epochs

B. Tokenization Examples

Example 1: Simple polymer

```
PSMILES: [*]CC(=O)OCC[*]

H-Net tokens:  [*] | CC | (=O) | OCC | [*]
SmilesPE:      [*] | CC(=O)O | CC | [*]
```

Example 2: Drug-like molecule

```
SMILES: CC(C)Cc1ccc(C(C)C(=O)O)cc1

H-Net tokens:  CC | (C) | Cc | 1 | ccc | (C | (C) | C(=O) | O) | cc | 1
SmilesPE:      CC(C) | Cc1ccc | (C(C) | C(=O)O) | cc1
```